

Crosscutting Areas

Spatio-Temporal Pricing for Ridesharing Platforms

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Abstract. Ridesharing platforms match drivers and riders to trips, using dynamic prices to balance supply and demand. A challenge is to set prices that are appropriately smooth in space and time, so that drivers with the flexibility to decide how to work will nevertheless choose to accept their dispatched trips rather than drive to another area or wait for higher prices or a better trip. In this work, we propose a complete information model that is simple yet rich enough to incorporate spatial imbalance and temporal variations in supply and demand—conditions that lead to market failures in today’s platforms. We introduce the *spatio-temporal pricing (STP) mechanism*. The mechanism is incentive-aligned, in that it is a subgame-perfect equilibrium for drivers to always accept their trip dispatches. From any history onward, the equilibrium outcome of the STP mechanism is welfare-optimal, envy-free, individually rational, budget balanced, and core-selecting. We also prove the impossibility of achieving the same economic properties in a dominant-strategy equilibrium. Simulation results show that the STP mechanism can achieve substantially improved social welfare and earning equity than a myopic mechanism.

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1. Introduction

Uber connected its first rider to a driver in San Francisco in the summer of 2010. Within one decade’s time, ridesharing platforms such as Uber and Lyft have radically changed the way people get around in urban areas. Comparing with traditional taxi systems, a distinct feature of ridesharing platforms is the emphasis on reliable transportation. For example, Uber’s mission is stated as “transportation as reliable as running water, everywhere, for everyone” (Uber 2016), and Lyft’s mission is “to provide the best, most reliable service possible by making sure drivers are on the road, when and where you need them most” (Lyft 2017). When demand exceeds supply, these platforms use dynamic surge pricing to guarantee that rider wait times would not exceed a few minutes (Rayle et al. 2014).

In addition to reliability for riders, the platforms also provide the flexibility for drivers to drive on their own schedule. Uber, for example, advertises itself as “work that put you first—drive when you want, earn what you need” (Uber 2017), and Lyft promises drivers “To drive or not to drive? It’s really up to you” (Lyft 2016). The real-time flexibility to decide when

and where to drive is an important reason that drivers drive for Uber (Hall and Krueger 2016) and substantially increases both driver supply and driver surplus in comparison with alternative, less flexible arrangements (Chen et al. 2019). More recently, Uber also started to provide drivers in some markets the option to accept only trips they want, based on the trip destinations and expected earnings (Uber 2019).

Despite their success, there remain a number of problems with the pricing and dispatching rules governing these ridesharing platforms, leading to various kinds of market failure and undercutting the endeavor to provide reliable yet flexible transportation. A particular concern is that trips may be mispriced relative to each other, incentivizing drivers to cherry pick (Marshall 2020, Cook et al. 2021). Drivers can also benefit from strategic behavior in the following scenarios, where there is spatial imbalance and temporal variation of supply and demand:

- (Spatial mispricing) When the price is substantially higher for trips that start in location *A* compared with an adjacent location *B*, drivers in location *B* that are close to the boundary can usefully decline trips. This spatial mispricing leads to drivers’ “chasing the

surge”—turning off a ridesharing app while relocating to another location (Campbell 2016, Chen 2016).

- (Temporal mispricing) When large events such as a sports game will soon end, drivers can anticipate that prices will increase substantially to balance supply and demand. In this case, many drivers around the stadium will decline trip dispatches and even go offline to wait in place (Gridwise 2017).

- (Network externalities) The origin-based surge pricing used by many platforms does not correctly factor market conditions at the destination of a trip. This incentivizes drivers to decline trips to destinations where the continuation payoffs are low, for example, quiet suburbs with low prices and long wait times (Paul 2018).

These pricing problems undercut the mission of reliable transport, with even high willingness-to-pay riders unable to get access to reliable service for certain trips, such as trips leaving the stadium before a game ends. These kinds of mispricing also make it difficult to accommodate drivers' idiosyncratic preferences, for example, over locations, because such features are used extensively by drivers to strategize for better earnings (Perea 2017). This can also lead to inequity, with demonstrated learning effects leading to differences in drivers' long-run earnings (e.g., a gender earnings gap; Cook et al. 2021), with potential consequences around driver churn from the platform.

Simple fixes by limiting dispatching transparency or drivers' flexibility are not fully effective. For example, when a platform hides trip destinations before the pick-up, experienced drivers will call riders to ask about trip details and cancel those trips that are not worthwhile (Cook et al. 2021). Nor does the imposition of penalties on drivers solve these problems, because drivers may decide to go offline or choose not to participate in the platform from certain locations or times.

We conceptualize many of the problems with current platforms as arising from prices failing to be appropriately “smooth” in space and time—if prices for trips are higher in one location, then they should be appropriately higher in adjacent locations; if demand would soon increase in a location, then the current prices should already be higher; and if destinations differ in continuation payoffs, then trip prices to these destinations need to reflect this. With appropriately smooth prices that correctly reflect the on-trip cost and network externality of completing each trip, drivers who retain the flexibility to decide how to work will still choose to accept any trip to which they are dispatched, providing reliability for riders. Many of the strategic behaviors on today's platforms are also symptoms of inefficiencies in dispatching, for example, when dispatching drivers to low-priced trips that send them away from a sports stadium five minutes

before a game ends. Correctly designed, ridesharing platforms can succeed in optimally orchestrating trips and providing reliable transportation for riders while leaving drivers with flexibility to decide how to work.

1.1. Our Results

In this work, we propose a framework for studying pricing and dispatching in the context of a ridesharing platform. The model that we introduce is simple but is the first in the literature to be rich enough to incorporate spatial imbalance and temporal variation of supply and demand—market conditions under which mispricing and strategic behavior arise in today's platforms. We propose the *spatio-temporal pricing* (STP) mechanism, which achieves the following properties that we consider very important for a sharing economy platform:

- **Welfare-optimality:** maximizing total rider values minus driver costs.
- **Incentive alignment:** the prices are appropriately smooth in space and time, such that drivers will always choose to accept any dispatched trips.
- **Robustness:** the mechanism updates the downstream plans after deviations from its dispatches.
- **Temporal consistency:** plans are computed and updated based on the current state but not past history, without using penalties or time-extended contracts.
- **Envy-freeness:** drivers at the same location and time do not envy each other's future payoffs; riders requesting the same trips do not envy each other's outcomes.
- **Core-selecting:** no coalition of riders and drivers can make a better plan among themselves.

Welfare-optimality and incentive alignment are necessary but are not enough to guarantee efficient and reliable operation. Robustness ensures the other properties from any point of time onward regardless of past deviations. This is important in the face of erroneous predictions, mistakes by participants, or unmodeled idiosyncratic preferences, and without robustness any solution would be brittle and poorly suited to practice. Temporal consistency matters, because using penalties (or threatening to fire drivers or shut down the system) is incompatible with the spirit of the sharing economy and the real-time flexibility of being able to choose how to work. Envy-freeness and core-selecting properties relate to fairness and to the long-run health of the marketplace. An envy-free mechanism removes unnecessary fluctuations in daily income that depend on lucky dispatches and reduces the long-run inequity in earnings from “learning-by-doing.” A core outcome ensures that no group of riders and drivers are disadvantaged, getting a worse outcome than the best plan they can form among themselves. In practice, a violation of the core makes a

platform vulnerable to a competitor that may take over such disadvantaged parts of the network.

The Model. We work in a complete information, discrete time, multiperiod, and multilocation model, addressing the challenge of promoting desirable behavior by drivers in the absence of time-extended contracts. At the beginning of each time period, based on the history, current positioning of drivers, and current and future demand, the STP mechanism dispatches each available driver to a rider trip, or to relocate, or to exit the platform for the planning horizon. The mechanism also determines a payment to be made if the driver follows the dispatch. Each driver then decides whether to follow the dispatch, or to decline and stay, or to relocate to any location, or to exit. After observing the driver actions in a period, the mechanism collects payments from the riders who are picked up and makes payments to the drivers who followed the dispatches. The main assumptions that we make are as follows:

- (i) Complete information about supply and demand over a finite planning horizon,
- (ii) Impatient, price-taking riders, with a value for being picked up at a particular time and location (and without preferences over drivers), and
- (iii) Drivers who each face the same costs for completing the same trip from some origin to some destination at some particular time (and without idiosyncratic preference over riders or locations), and who are willing to provide trips until the end of the planning horizon.

We do allow for heterogeneity in rider values and trip details (the origin, destination, and time of a trip). For drivers, we allow them to become available at different times and locations, and we also model the distinction between a driver who is already driving in the platform (e.g., finishing a trip), and a driver who has not yet started driving and thus needs to make an entry decision (e.g., dropping off a child at school at a specific location and time and willing to drive afterward). We also allow a driver who is asked to exit the platform earlier than their intended exit time to incur a one-time exit cost, modeling the forgone opportunity of outside options after the driver has been driving in the platform for some time.

Main Results. We first show that the welfare-optimal planning problem can be reduced to a minimum cost flow (MCF) problem. The integrality of the linear program (LP) of the MCF guarantees the existence of anonymous (not depending on the identity of rider or driver), origin-destination, competitive equilibrium (CE) prices, allowing the price of a trip to depend on market conditions at both the origin and destination. The lattice structure of the dual LP also implies that drivers' total utilities among all CE plans form a lattice. The STP mechanism uses *driver-pessimal*

CE prices, computing a driver-pessimal CE plan at the beginning of the planning horizon, as well as after any deviations from the current plan. This induces an extensive-form game among the drivers, where the total payoff to each driver is determined by the mechanism's dispatch and payment rules, as well as the actions taken by the other drivers. The main result is that the STP mechanism satisfies all the desiderata outlined previously.

The STP mechanism uses driver-pessimal rather than driver-optimal CE prices. The driver-optimal analog of the STP mechanism reflects the payments of a Vickrey-Clarke-Groves (VCG) mechanism but fails to align incentives (see Example 3). In a driver-pessimal plan, a driver's continuation payoff from some location and time onward is equal to the welfare gain in the economy from adding an additional driver at this location and time. The M^i concavity of the MCF problems (Murota 2003) implies a stronger substitution between drivers at the same location at the same time (than between drivers at different locations or times), allowing us to reason about the welfare gains and prove that accepting the mechanism's dispatches forms a subgame-perfect equilibrium. Under a driver-optimal plan, each driver's continuation payoff is equal to the welfare-loss in the economy from losing the driver. This "marginal product" can increase over time, as the set of trips that can be completed by the rest of the drivers becomes smaller. This breaks incentives so that a driver can usefully deviate from a suggested dispatch and trigger a plan update.

We also prove an impossibility result that no dominant-strategy mechanism has the same economic properties. For three stylized scenarios (the end of an event, the morning rush hour, and trips to and from the airport with unbalanced flows), we compare the STP mechanism with a *myopic pricing mechanism* that simply clears the market for each location at each time, without taking future demand or supply into consideration. Simulation results (Online Appendix A) suggest that STP achieves substantially higher social welfare and highlight the failure of incentive alignment and the large variance in driver earnings because of nonsmooth prices in myopic mechanisms.

The main operational insight from this paper is that each trip should be priced (as it is under the STP mechanism) at the welfare contribution of an extra driver at the origin of the trip, minus the welfare contribution of an extra driver at the destination of the trip, plus the cost to a driver to complete this trip. This correctly "prices in" the externality imposed on the system by moving a driver from the origin to the destination. As an example, an extra driver at a stadium five minutes before a game ends is very valuable, so with this approach to pricing, the price will go up smoothly before a game ends, providing riders with

high willingness-to-pay reliable access to transportation. Prices determined in this way will also be smooth in space, reducing drivers' incentives to chase the surge. From this insight comes the importance to practice of estimating the welfare contribution of an extra driver, likely the expected welfare contribution in an actual deployment, using data from the current, typically suboptimal, operations.

1.2. Related Work

Banerjee et al. (2015) adopt a queuing-theoretic approach in analyzing the effect of dynamic pricing on the revenue and throughput of ridesharing platforms, assuming stationary system state. When the platform correctly estimates supply and demand, the optimal dynamic pricing strategy does not achieve better performance than the optimal static pricing strategy. However, dynamic pricing is more robust to fluctuations and to mis-estimation of system parameters.

By analyzing a continuum model with stationary demand and unlimited driver supply at fixed costs, Bimpikis et al. (2019) show that a platform's profit is maximized when the demand pattern across locations is balanced. They show in simulation that in comparison with a single price, origin-based pricing improves profit, although there is not a substantial gain from using origin-destination based pricing. Our model is quite distinct in that it is not a continuum or stationary model, does not have unlimited driver supply, and is focused on welfare. Banerjee et al. (2017) model a shared vehicle system as a continuous-time Markov chain and establish approximation guarantees for a static, state-independent policy with respect to (w.r.t.) the optimal, state-dependent policy.

Castillo et al. (2017) study the "wild goose chase" phenomena in platforms that myopically dispatch the closest drivers to rider requests. When demand much exceeds supply, drivers spend too much time driving to pick up riders, leading to long wait times and decreased welfare and revenue. Under a model with stationary supply and demand, the authors establish the importance of dynamic pricing for these myopic dispatching schemes, in keeping enough open cars to avoid inefficient long pick-ups. Yan et al. (2020) provide a review of matching and dynamic pricing in ridesharing platforms. Garg and Nazerzadeh (2020) study a dynamic stochastic model and show that when the surge pricing needs to be origin-based and cannot depend on the length of the trip, additive surge is more incentive compatible for drivers in practice than is multiplicative surge.

There are various empirical studies of the Uber platform as a two-sided marketplace (Cohen et al. 2016, Hall and Krueger 2016, Hall et al. 2017), analyzing the labor market of Uber's drivers, the longer-term labor market equilibration, and consumer surplus. By

analyzing drivers' hourly earnings, Chen et al. (2019) show that drivers' reservation wages vary significantly over time, and that the real-time flexibility of being able to choose when to work increases driver surplus and driver supply. In regard to dynamic pricing, Chen and Sheldon (2015) show that surge pricing increases the supply of drivers at times when the surge pricing is high, and Lu et al. (2018) show that surge pricing incentivizes drivers to relocate to higher surge areas. A case study into an outage of Uber's surge pricing during the 2014–2015 New Year's Eve found a large increase in riders' waiting times and a large decrease in the percentage of requests completed (Hall et al. 2015).

In this work, we made use of the connections between LP duality and market equilibrium to prove the existence and structure of CE. These connections are widely applied in many matching and assignment settings (Shapley and Shubik 1971, Parkes and Ungar 2000, Bikhchandani and Ostroy 2002). The existing literature, however, does not provide a framework to study the dynamic incentive properties in our setting in regard to drivers' decisions about accepting dispatches.

In combinatorial auctions, the connection between LP duality and CE allows the design of iterative auctions that can be interpreted as primal-dual algorithms that solve the optimal allocation problem (Parkes and Ungar 2000, Mishra and Parkes 2007). The challenge is to elicit valuation functions, and these designs induce an extensive-form game among agents who participate in each round of the auction, and before an allocation is determined. This is different from our setting, where the challenge is not asymmetric information but rather agency, and where drivers' strategic behavior is realized after a plan has been computed.

The *matching with substitutes* literature (Kelso and Crawford 1982, Gul and Stacchetti 1999) establishes the existence and lattice structure of CE payoffs for two-sided matching where the preference of an agent over the agents on the other side satisfy the *gross substitutes* (GS) condition. Our problem cannot be considered as two-sided matching with substitutes, because from a driver's perspective, riders on different segments of the same path may be complements to each other.

The literature on *trading networks* studies economic models where agents in a network can transact via bilateral contracts (Ostrovsky 2008; Hatfield et al. 2013, 2021). A CE exists when preferences satisfy a *full substitution* property, and the utilities of agents on either end of an acyclic network forms a lattice. Although the optimal planning problem we study can be reduced to a trading network, where drivers and riders trade the right to use a car for the rest of the planning horizon, it is not apparent how to use this

mapping to establish the incentive properties of a ridesharing mechanism (see Online Appendix E.2).

Dynamic variations of the VCG mechanism (Cavallo et al. 2009, Bergemann and Välimäki 2010, Athey and Segal 2013) are able to truthfully implement efficient decision policies, where agents receive private information over time. These mechanisms are not suitable for our problem, because some drivers may be paid negative payments for certain periods of time, which would lead them to decline dispatches. See Online Appendix E.1 for examples and detailed discussions.

Principal-agent problems are, of course, studied extensively in contract theory (Bolton and Dewatripont 2005, Salanié 2005). When information asymmetry arises before the time of contracting, with private information, this is a problem of *adverse selection*. When information asymmetry arises after the time of contracting, through hidden actions, this is a problem of *moral hazard*. Where contracts cannot be perfectly enforced, *relational incentive contracts* (Levin 2003) are self-enforcing by threatening to terminate an agent following poor performance. In our model, there is neither private information nor hidden actions. Rather, the challenge that we face is one of incentive alignment in the absence of time-extended contracts, so that drivers retain the flexibility to decide on actions without incurring penalties or facing termination threats.

2. Preliminaries

Let T be the length of the planning horizon, starting at time $t = 0$ and ending at time $t = T$. We adopt a discrete time model, refer to each time point t as “time t ,” and call the duration between time t and time $t + 1$ a *time period*. We may think about each time period as approximately five minutes, and with $T = 6$, the planning horizon would be half an hour. Trips start and end at time points. Denote $[T] = \{0, 1, \dots, T\}$ and $[T - 1] = \{0, 1, \dots, T - 1\}$.

Let $\mathcal{L} = \{A, B, \dots\}$ be a set of $|\mathcal{L}|$ discrete locations, and we adopt a and b to denote generic locations. For all $a, b \in \mathcal{L}$ and $t \in [T]$, the triple (a, b, t) denotes a *trip* with origin a , destination b , starting at time t . Each trip can represent (i) taking a rider from a to b at time t , (ii) relocating without a rider from a to b at time t , and (iii) staying in the same location for one period of time (in which case $a = b$). Let the distance $\delta: \mathcal{L} \times \mathcal{L} \rightarrow \mathbb{N}$ be the number of time periods needed to travel between locations, so that trip (a, b, t) ends at $t + \delta(a, b)$. We allow $\delta(a, b) \neq \delta(b, a)$ for locations $a \neq b$, modeling asymmetric traffic flows. We can also allow the distance between a pair of locations to change over time, modeling the changes in traffic conditions, that is, a trip from a to b starting at time t ends at time $t + \delta(a, b, t)$. This does not affect the results, and we

keep $\delta(a, b)$ for simplicity of notation. We assume $\delta(a, b) \geq 1$ for all $a, b \in \mathcal{L}$, and $\delta(a, a) = 1$ for all $a \in \mathcal{L}$. Set $\mathcal{T} \triangleq \{(a, b, t) \mid a \in \mathcal{L}, b \in \mathcal{L}, t \in \{0, 1, \dots, T - \delta(a, b)\}\}$ denotes the set of all feasible trips within the planning horizon.

Let $\mathcal{D}$ denote the set of drivers, with $m \triangleq |\mathcal{D}|$. Each driver $i \in \mathcal{D}$ is characterized by *type* $\theta_i = (\beta_i, \ell_i, \underline{\tau}_i, \bar{\tau}_i)$ —driver i is able to enter the platform at location ℓ_i and time $\underline{\tau}_i$, and plans to exit the platform at time $\bar{\tau}_i$ (with $\underline{\tau}_i < \bar{\tau}_i$). β_i indicates driver i 's entrance status. A driver with $\beta_i = 0$ has not yet entered the platform, and needs to make an entry decision—consider a driver who is willing to drive after dropping her daughter at school at location ℓ_i at time $\underline{\tau}_i$. A driver with $\beta_i = 1$ has already entered the platform (she may be completing an earlier trip, or relocating to another location), and will become available to pick up again at $(\ell_i, \underline{\tau}_i)$. Here we make the assumption (S1) that *driver types are known to the mechanism and that all drivers stay until at least the end of the planning horizon, and do not have a preference over riders or location, including where they finish their last trip in the planning horizon*.

A driver who completes a trip $(a, b, t) \in \mathcal{T}$ incurs a cost $c_{a,b,t} \geq 0$, which models the cost of time, driving, fuel, wear-and-tear, and so on. A driver who has already entered the platform may exit earlier than her intended exit time, in which case she will not be able to complete any trip in the remaining periods of this planning horizon. Exiting Δ periods earlier than time T incurs a one-time cost of $\kappa_\Delta \geq 0$ (with $\kappa_0 = 0$), modeling the forgone opportunity of outside employment options, after driving for the platform for some time. A driver with $\beta_i = 0$ who does not enter the platform at $(\ell_i, \underline{\tau}_i)$ does not incur any cost, and will not enter at a later time. Drivers have quasi-linear utilities and seek to maximize the total payments received over the planning horizon minus the total costs.

Denote $\mathcal{R}$ as the set of riders, each intending to take a single trip during the planning horizon. The *type* of rider $j \in \mathcal{R}$ is (o_j, d_j, τ_j, v_j) , where o_j and d_j are the trip origin and destination, τ_j the requested start time, and $v_j \geq 0$ the *value* for the trip. The value v_j models the rider's willingness-to-pay over and above a base payment that covers the additional cost that a driver incurs for picking up a rider in comparison with just relocating (e.g., extra wear-and-tear, loss of privacy, inconvenience). This base payment is always made for a matched trip and allows us to model a driver's cost as depending on the origin, destination, and time of a trip, and irrespective of whether there is a passenger in the car. With this, the prices we determine are the amount to pay on top of the base amount. We assume (S2) that *riders are impatient, only value trips starting at τ_j , are not willing to relocate or walk from a drop-off point to their actual, intended destination, and do not*

have preference over drivers. Rider utility is quasi-linear, with utility $v_j - p$ to rider j for a trip at (incremental to base) price p .

We assume the platform has complete information about supply and demand over the planning horizon (travel times, trip costs, driver and rider types, including driver entry during the planning horizon). We assume drivers have the same information and that this is common knowledge among drivers (more generally, it is sufficient that it be common knowledge among drivers that the platform has the correct information). Unless otherwise noted, we assume properties (S1), (S2), and complete, symmetric information throughout the paper. Detailed discussions on the effect of relaxing these assumptions are provided in Section 5.

At each time t , a driver is *en route* if she started her last trip from a to b at time t' (with or without a rider), and $t < t' + \delta(a, b)$. A driver is *available* if she has entered or is able to enter the platform, has not yet exited, and is not *en route*. A driver who is available at time t and location a is able to complete a pick-up at this location and time. We allow a driver to drop-off a rider and pick-up another rider in the same location at the same time point (see Online Appendix B).

A *path* is a sequence of tuples (a, b, t) , representing driver entrance, exit, and the trips she takes over the planning horizon. Let Z_i denote the set of all *feasible paths* of driver i , with $Z_{i,k} \in Z_i$ to denote her k^{th} feasible path. The path $Z_{i,0}$ includes no trip: for a driver with $\beta_i = 0$, $Z_{i,0}$ models the option to not enter the platform at all; for a driver such that (s.t.) $\beta_i = 1$, $Z_{i,0}$ models the option exit immediately at (ℓ_i, τ_i) . For each $k = 1, \dots, |Z_i|$, $Z_{i,k}$ is a path that starts at (ℓ_i, τ_i) , with the starting time and location of each successive trip equal to the ending time and location of the previous trip. Denote $(a, b, t) \in Z_{i,k}$ if path $Z_{i,k}$ includes (or *covers*) trip (a, b, t) , and let $\lambda_{i,k}$ be the total cost of the k^{th} path to driver i . We know that $\lambda_{i,0} = 0$ if $\beta_i = 0$, $\lambda_{i,0} = \kappa_{T-\tau_i}$ if $\beta_i = 1$, and for $k \geq 1$, $\lambda_{i,k} = \sum_{(a,b,t) \in Z_{i,k}} c_{a,b,t} + \kappa_\Delta$, if path $Z_{i,k}$ ends Δ periods earlier than T .

Driver i who takes the path $Z_{i,k}$ is able to pick up rider j if $(o_j, d_j, \tau_j) \in Z_{i,k}$; however, a path specifies only the movement in space and time, and does not specify whether a rider is picked up for each of the trips on the path. Let an *action path* for driver i be a sequence of tuples, each of them can either be of the form (a, b, t) , representing a relocation trip from a to b at time t without a rider, or be of the form (a, b, t, j) , in which case the driver sends rider j from a to b at time t (thus requiring $(a, b, t) = (o_j, d_j, \tau_j)$). Let $\tilde{Z}_i$ be the set of all feasible action paths of driver i (the feasibility of an action path is similar to that of a path). For an action path $\tilde{z}_i \in \tilde{Z}_i$, denote $(a, b, t) \in \tilde{z}_i$ or $(a, b, t, j) \in \tilde{z}_i$ if the action path includes a relocation or rider trip from a to b at time t . A driver taking action path $\tilde{z}_i$ that is

consistent with path $Z_{i,k}$ (i.e., results in the same movement in space and time) incurs a total cost of $\lambda_{i,k}$.

Example 1. The planning horizon is $T = 2$ and there are two locations $\mathcal{L} = \{A, B\}$ with distance $\delta(A, A) = \delta(B, B) = 1$ and $\delta(A, B) = \delta(B, A) = 2$ (Figure 1). Trip costs are two per period of time, that is, $c_{a,b,t} = 2\delta(a, b)$ for all $(a, b, t) \in \mathcal{T}$, and the opportunity cost of exiting early is $\kappa_\Delta = \Delta$. There is one driver, who has not yet entered the platform (i.e., $\beta_1 = 0$) but is able to enter at time $\tau_1 = 0$ at location $\ell_1 = A$, and plans to leave at time $\bar{\tau}_1 = 2$. There are three riders:

- Rider 1: $o_1 = A, d_1 = A, \tau_1 = 0, v_1 = 5$,
- Rider 2: $o_2 = A, d_2 = A, \tau_2 = 1, v_2 = 6$, and
- Rider 3: $o_3 = A, d_3 = B, \tau_3 = 0, v_3 = 8$.

In addition to not entering the platform at all, which corresponds to path $Z_{1,0}$ with cost $\lambda_{1,0} = 0$, there are three more feasible paths for driver 1: $Z_{1,1} = ((A, A, 0), (A, A, 1))$, $Z_{1,2} = ((A, B, 0), (A, B, 1))$, and $Z_{1,3} = ((A, A, 0), (A, B, 1))$. In $Z_{1,3}$, the driver exits one period before the end of planning horizon. The path costs are $\lambda_{1,1} = c_{A,A,0} + c_{A,A,1} = 4$, $\lambda_{1,2} = c_{A,B,0} + c_{A,B,1} = 4$, and $\lambda_{1,3} = c_{A,A,0} + \kappa_1 = 3$. Path $((A, A, 0), (A, B, 1))$ is infeasible, because the last trip ends later than the driver's leaving time. Similarly, paths $((A, B, 0), (B, B, 1))$ and $((A, A, 0), (B, B, 1))$ are infeasible.

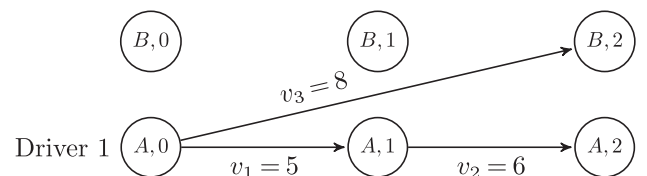
In addition to not entering, there are eight feasible actions paths of driver 1: $((A, B, 0))$, relocating from A to B at time 0, and $((A, B, 0, 3))$, sending rider 3 from A to B at time 0, are both consistent with the path $Z_{1,2}$, and both have cost 4. Four action paths, $((A, A, 0), (A, A, 1))$, $((A, A, 0, 1), (A, A, 1, 1))$, $((A, A, 0), (A, A, 1, 2))$, $((A, A, 0, 1), (A, A, 1, 2))$, are consistent with $Z_{1,1}$ and have cost 4. Both $((A, A, 0))$ and $((A, A, 0, 1))$ are consistent with $Z_{1,3}$ and have cost 3.

We now provide an informal timeline of a *ridesharing mechanism* (see Section 4 for a formal definition). At each time point $t \in [T - 1]$, given the history of trips, current positioning and availability of drivers, and current and future driver supply and rider demand for trips:

1. The ridesharing mechanism determines for each rider with trip start time t , whether a driver will be dispatched to pick her up, and if so, the price of her trip.

2. The mechanism dispatches available drivers to pick up riders, to relocate, or to exit (for drivers already in the platform), or not to enter (for drivers who have

Figure 1. Economy in Example 1, with Two Locations A, B , Two Time Periods, and Three Riders



not entered, with $\underline{\tau}_i = t$ and $\beta_i = 0$). The mechanism also determines the payments offered to drivers for accepting the dispatches.

3. Each available driver decides whether to accept the dispatch, or to deviate and either stay in the same location, or relocate, or exit/not enter. A driver may still decide to enter the platform even if asked not to do so. The mechanism collects and makes payments based on driver actions.

Any undischpatched, available driver makes their own choices of actions. We assume that any driver already *en route* will continue their current trip. A driver's payment in a period in which the driver declines a dispatch is zero, so that drivers are not charged penalties for deviation.

As a baseline, we define the following myopic pricing mechanism. For each rider $j \in \mathcal{R}$, denote the per-period surplus of her trip as $w_j \triangleq (v_j - c_{o_j, d_j, \tau_j}) / \delta(o_j, d_j)$.

Definition 1 (Myopic Pricing Mechanism). At each time point $t \in [T-1]$, for each location $a \in \mathcal{L}$, the *myopic pricing mechanism* dispatches available drivers at (a, t) to riders with $(o_j, \tau_j) = (a, t)$ and $w_j \geq 0$, in decreasing order of w_j . The mechanism sets a market clearing rate $\rho_{a,t}$ (i.e., between highest unallocated w_j and lowest allocated w_j), and sets prices $p_{a,b,t} = \delta(a, b)\rho_{a,t} + c_{a,b,t}$ for each destination $b \in \mathcal{L}$, which is offered to all dispatched drivers and collected from all riders who are picked up.

The market clearing prices may not be unique, and a fully defined myopic mechanism must provide a rule for picking a particular set of prices. This mechanism has anonymous, origin-based pricing, and is very simple in ignoring the need for smooth pricing, or future supply and demand. But its simplicity means that it fails to optimize social welfare or to set prices that are spatially and temporally smooth, leading to various market failures.

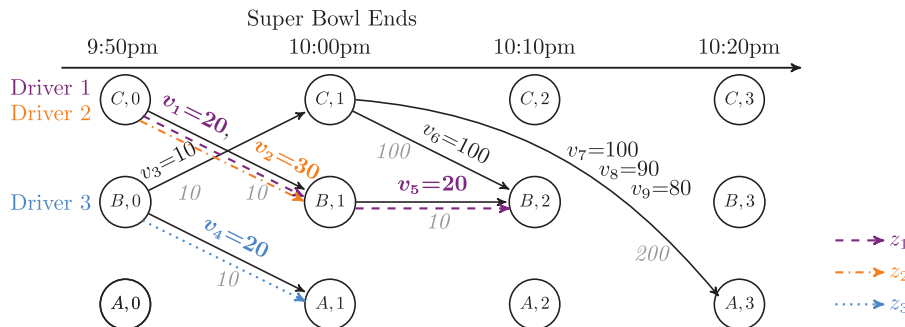
Example 2 (Super Bowl Example). Consider the economy illustrated in Figure 2, modeling the end of a

sports event, with $T = 3$ time periods. Each time period is 10 minutes. Time $t = 0$ is 9:50 p.m. and 10 minutes before the game ends. There are three locations A , B , and C with symmetric distances $\delta(A, A) = \delta(B, B) = \delta(C, C) = \delta(A, B) = \delta(B, A) = \delta(B, C) = \delta(C, B) = 1$ and $\delta(A, C) = \delta(C, A) = 2$. Drivers 1 and 2 enter at location C at time 0, whereas driver 3 enters at B at time 0, with exit times $\bar{\tau}_i = T$ for all $i \in \mathcal{D}$. Riders' trips and values are as shown in the figure. The game ends at location C at time 1, where many riders with high values will request rides. Trips cost 10 per time period, that is, $c_{a,b,t} = 10\delta(a, b)$, $\forall (a, b, t) \in \mathcal{T}$, and early exiting costs are $\kappa_\Delta = 5\Delta$.

Suboptimal Welfare. Under the myopic pricing mechanism, at time 0, drivers 1 and 2 are dispatched to pick up riders 1 and 2, respectively, and driver 3 is dispatched to pick up rider 4. At time 1, driver 1 picks up rider 5. Assuming optimal exiting (i.e., driver 1 exits at time 2 at a cost of $\kappa_1 = 5$, whereas drivers 2 and 3 exit at time 1 and each incurs an exit cost of $\kappa_2 = 10$), the total social welfare achieved under the myopic pricing mechanism is only $v_1 + v_2 + v_4 + v_5 - 10 \times 4 - 5 - 10 - 10 = 25$. This also illustrates that the myopic pricing mechanism does not achieve any constant fraction of the optimal welfare.

Unsmooth Prices in Space and Time. The set of market clearing rates for C at time 0 is $\rho_{C,0} \in [0, w_1] = [0, 10]$; thus, the possible market clearing prices for the trip $(C, B, 0)$ is $p_{C,B,0} \in [10, 20]$. The price for the $(B, B, 1)$ trip is $p_{B,B,1} = c_{B,B,1} = 10$, because there is excess supply. At time 1, because no driver is able to pick up the four riders at location C , the lowest market clearing rate is $\rho_{C,1} = w_6 = 90$. The prices therefore must be at least $p_{C,B,1} \geq \delta(C, B)\rho_{C,1} + c_{C,B,1} = 100$ and $p_{C,A,1} \geq \delta(C, A)\rho_{C,1} + c_{C,A,1} = 200$. The set of lowest market clearing prices are shown in italics in Figure 2, below the edges corresponding to the trips. We can see that the price for the (C, B) trip jumps from 10 to 100 within one period of time. Moreover, there exist large

Figure 2. (Color online) Super Bowl Game: Time 0 Plan Under the Myopic Pricing Mechanism



Notes. Color-coded paths $\{z_i\}_{i \in \mathcal{D}}$ and rider values in bold indicate the movement of drivers in space and time, as well as the riders picked up by each driver. Numbers in italics below trips is the set of lowest origin-based market clearing prices.

gaps between prices for trips originating from neighboring locations (compare, e.g., $p_{C,B,1} = 100$ and $p_{B,B,1} = 10$).

Incentivizing Strategic Behavior. Because $p_{C,B,0} \leq 20$, the highest possible total utility to driver 1 under any myopic pricing mechanism would be $(20 - 10) + (10 - 10) - 5 = 5$, and the utility to driver 2 will not exceed $(20 - 10) - 10 = 0$. When all drivers follow the dispatches, the outcome fails to be envy-free because the two drivers who start at the same location and time have different total payoffs.

Now suppose driver 1 deviates from the dispatch and stays in C until time 1. The mechanism would then dispatch her to pick up rider 6, and driver 1 would be paid the new market clearing price of at least $\delta(C, B)w_7 + c_{C,B,1} = 50$. This is a useful deviation, because by exiting at time 2, her utility is now at least $-10 + (50 - 10) - 5 = 25$. Also observe that by pricing in a myopic manner, strategic drivers are rewarded substantially higher earnings than drivers who are straightforward and accept all dispatches.

3. Static CE Mechanism

In this section, we formulate the welfare-optimal planning problem, define CE prices, and prove a welfare theorem, the core equivalence, and a lattice structure of drivers' utilities among all CE plans.

3.1. Plans

A *plan* describes the paths taken by all drivers until the end of the planning horizon, rider pick-ups, and payments for riders and drivers for each trip associated with these paths.

Formally, a plan is the four-tuple $(x, \tilde{z}, q, r)$, where x is the indicator of rider pick-ups, that is, for all riders $j \in \mathcal{R}$, $x_j = 1$ if rider j is picked up according to the plan, and $x_j = 0$ otherwise; $\tilde{z}$ is a vector of action paths, where $\tilde{z}_i \in \tilde{\mathcal{Z}}_i$ is the dispatched action path taken by driver i ; q_j denotes the payment made by rider j ; $r_{i,t}$ denotes the payment made to driver i at time t ; and let $r_i \triangleq \sum_{t=0}^T r_{i,t}$ denote the total payment to driver i . If $\tilde{z}_i$ is consistent with $Z_{i,k}$, the k^{th} feasible path of driver i , then the driver incurs a total cost of $\lambda_{i,k}$, and her utility is $\pi_i \triangleq r_i - \lambda_{i,k}$.

A plan $(x, \tilde{z}, q, r)$ is *feasible* if for each rider $j \in \mathcal{R}$, $x_j = \sum_{i \in \mathcal{D}} \mathbb{1}\{(o_j, d_j, \tau_j, j) \in \tilde{z}_i\} \in \{0, 1\}$, where $\mathbb{1}\{\cdot\}$ is the indicator function. Unless stated otherwise, when we mention a plan in the rest of the paper, it is assumed to be feasible. For the *budget balance* (BB) of a plan, we need $\sum_{j \in \mathcal{R}} q_j \geq \sum_{i \in \mathcal{D}} r_i$, with strict budget balance if the inequality holds with equality. A plan is *individually rational for riders* if $x_j v_j \geq q_j$, $\forall j \in \mathcal{R}$. A plan is *individually rational for drivers* if $\pi_i \geq 0$ for all $i \in \mathcal{D}$ s.t. $\beta_i = 0$, that is, drivers that are not yet in the platform do not

get negative utility from participating. A plan is *envy-free for riders* if no rider strictly prefers the outcome of another rider requesting the same trip, that is, $x_j v_j - q_j \geq x_{j'} v_j - q_{j'}$ for all $j, j' \in \mathcal{R}$ s.t. $o_j = o_{j'}$, $d_j = d_{j'}$, and $\tau_j = \tau_{j'}$. A plan is *envy-free for drivers* if any pair of drivers with the same type have the same utility: $\pi_i = \pi_{i'}$ for all $i, i' \in \mathcal{D}$ s.t. $\underline{\tau}_i = \underline{\tau}_{i'}$, $\ell_i = \ell_{i'}$, and $\beta_i = \beta_{i'}$.

Plans with a particular kind of anonymity structure can also be defined by associating prices with every possible trip.

Definition 2 (Anonymous Trip Prices). A plan $(x, \tilde{z}, q, r)$ uses *anonymous trip prices* if there exist prices $p = \{p_{a,b,t}\}_{(a,b,t) \in \mathcal{T}}$ such that for all $(a, b, t) \in \mathcal{T}$, we have the following:

- (i) all riders taking the same (a, b, t) trip are charged the same payment $p_{a,b,t}$, and there is no payment by riders who are not picked up, and
- (ii) all drivers that are dispatched on a rider trip from a to b at time t are paid the same amount $p_{a,b,t}$ for the trip at time t , and there is no other payment to or from any driver.

Given dispatches $(x, \tilde{z})$ and anonymous trips prices p , all payments are fully determined: the total payment to driver i is $r_i = \sum_{j \in \mathcal{R}} \mathbb{1}\{(o_j, d_j, \tau_j, j) \in \tilde{z}_i\} p_{o_j, d_j, \tau_j}$ and the payment made by rider j is $q_j = x_j p_{o_j, d_j, \tau_j}$. For this reason, we will represent plans with anonymous trip prices as $(x, \tilde{z}, p)$. By construction, plans with anonymous trip prices are strictly budget balanced.

Definition 3 (Competitive Equilibrium). A plan with anonymous trip prices $(x, \tilde{z}, p)$ forms a CE if

- (i) (rider best response) all riders $j \in \mathcal{R}$ that can afford the ride are picked up, i.e. $v_j > p_{o_j, d_j, \tau_j} \Rightarrow x_j = 1$, and all riders that are picked up can afford the price: $x_j = 1 \Rightarrow v_j \geq p_{o_j, d_j, \tau_j}$, and

- (ii) (driver best response) $\forall i \in \mathcal{D}$, $\pi_i = \max_{k=0, \dots, |\mathcal{Z}_i|} \left\{ \sum_{(a,b,t) \in \mathcal{Z}_{i,k}} \max\{p_{a,b,t}, 0\} - \lambda_{i,k} \right\}$, that is, each driver achieves the highest possible utility given prices and the set of feasible paths.

Given any set of anonymous trip prices p , let anonymous trip prices p^+ be defined as $p_{a,b,t}^+ \triangleq \max\{p_{a,b,t}, 0\}$ for each $(a, b, t) \in \mathcal{T}$.

Lemma 1. *Given any CE plan $(x, \tilde{z}, p)$, the plan with anonymous prices $(x, \tilde{z}, p^+)$ also forms a CE, and has the same driver and rider payments and utilities as those under $(x, \tilde{z}, p)$.*

The lemma implies that when studying the set of possible rider and driver payments and utilities among all CE outcomes, it is without loss to consider only anonymous trip prices that are nonnegative. We leave the proof of this lemma to Online Appendix C.1. Intuitively, driver and rider best response requires that prices must be nonnegative for any trip that is requested by any rider. Thus, changing prices from p to

p^+ does not affect the payments for any rider or driver, or the best response on the riders' side. The driver best response property also continues to hold, because $\max\{p_{a,b,t}, 0\} = \max\{p_{a,b,t}^+, 0\}$ for all $(a, b, t) \in \mathcal{T}$.

For a given mechanism that dispatches all available drivers at all times, and with knowledge of supply and demand and assuming drivers follow suggested dispatches, we can compute the outcome through the planning horizon. We call this the “time 0 plan,” which consists of the assignments of riders, the action paths taken by drivers, and the payment schedule.

3.2. Optimal Plans and CE Prices

The welfare-optimal planning problem can be formulated as an integer linear program (ILP) that determines rider pick-ups and driver paths, followed by an assignment of riders to drivers whose paths cover the rider trips. Let x_j be the indicator that rider $j \in \mathcal{R}$ is picked up, and $y_{i,k}$ be the indicator that driver i takes $Z_{i,k}$, her k^{th} feasible path in $\mathcal{Z}_i$. Relaxing the integrality constraints, the LP relaxation of the welfare-optimal planning program is as follows:

$$\max_{x,y} \sum_{j \in \mathcal{R}} x_j v_j - \sum_{i \in \mathcal{D}} \sum_{k=0}^{|\mathcal{Z}_i|} y_{i,k} \lambda_{i,k}, \quad (1)$$

$$\text{s.t. } \sum_{j \in \mathcal{R}} x_j \mathbb{1}\{(o_j, d_j, \tau_j) = (a, b, t)\}$$

$$\leq \sum_{i \in \mathcal{D}} \sum_{k=0}^{|\mathcal{Z}_i|} y_{i,k} \mathbb{1}\{(a, b, t) \in Z_{i,k}\}, \quad \forall (a, b, t) \in \mathcal{T}, \quad (2)$$

$$\sum_{k=0}^{|\mathcal{Z}_i|} y_{i,k} = 1, \quad \forall i \in \mathcal{D}, \quad (3)$$

$$x_j \leq 1, \quad \forall j \in \mathcal{R}, \quad (4)$$

$$x_j \geq 0, \quad \forall j \in \mathcal{R}, \quad (5)$$

$$y_{i,k} \geq 0, \quad \forall i \in \mathcal{D}, k = 1, \dots, |\mathcal{Z}_i|. \quad (6)$$

We refer to (1) as the primal LP. Constraint (3) requires that each driver takes exactly one path (which includes the path $Z_{i,0}$ representing not entering/exiting immediately). The feasibility constraint (2) requires that for each trip $(a, b, t) \in \mathcal{T}$, the number of riders who request this trip and are picked up is no greater than the total number of drivers whose paths cover this trip. Once the rider pick-ups x and driver paths y are computed, (2) guarantees that each rider with $x_j = 1$ can be assigned to a driver. The constraint $y_{i,k} \leq 1$, that each path is taken by each driver at most once is guaranteed by imposing (3) and (6) and is omitted.

Lemma 2 (Integrality). *There exists an integer optimal solution to the LP (1).*

We leave the proof of this lemma to Online Appendix C.2, showing there is a correspondence to a minimum cost flow (MCF) problem (it is common for

transportation problems to be formulated as network flow problems; Ahuja et al. 1993). In the MCF, drivers flow through a network with vertices corresponding to (location, time) pairs, edges corresponding to trips, and with edge costs equal to driver's costs minus riders' values. The MCF has integral optimal solutions because of total unimodularity, and this reduction can also be used to efficiently solve for the optimal plans.

Let $p_{a,b,t}$, π_i , and u_j denote the dual variables corresponding to the primal constraints (2), (3), and (4), respectively. The dual LP of (1) is as follows:

$$\min \sum_{i \in \mathcal{D}} \pi_i + \sum_{j \in \mathcal{R}} u_j, \quad (7)$$

$$\text{s.t. } \pi_i \geq \sum_{(a,b,t) \in Z_{i,k}} p_{a,b,t} - \lambda_{i,k}, \quad \forall k = 0, 1, \dots, |\mathcal{Z}_i|, \quad \forall i \in \mathcal{D}, \quad (8)$$

$$u_j \geq v_j - p_{o_j, d_j, \tau_j}, \quad \forall j \in \mathcal{R}, \quad (9)$$

$$p_{a,b,t} \geq 0, \quad \forall (a, b, t) \in \mathcal{T}, \quad (10)$$

$$u_j \geq 0, \quad \forall j \in \mathcal{R}. \quad (11)$$

Lemma 3 (Welfare Theorem). *A dispatching $(x, \tilde{z})$ is welfare-optimal if and only if there exists anonymous trip prices p s.t. the plan $(x, \tilde{z}, p)$ forms a competitive equilibrium. Such CE plans always exist and are efficient to compute. Moreover, these plans are strictly budget balanced and are individually rational and envy-free for both riders and drivers.*

See Online Appendix C.3 for the proof of this lemma. Briefly, the dual variables π and u can be interpreted as the utilities of drivers and riders, when the anonymous trip prices are given by p . We then make use of Lemma 1 and the standard observations about complementary slackness conditions and their connection with competitive equilibria (Bertsekas 1990, Parkes and Ungar 2000). By integrality, CE plans always exist and can be efficiently computed by solving the primal and dual LPs of the MCF problem.

For two driver utility profiles $\pi = (\pi_1, \dots, \pi_m)$, $\pi' = (\pi'_1, \dots, \pi'_m)$ that correspond to CE plans, let the *join* $\bar{\pi} = \pi \vee \pi'$ and the *meet* $\underline{\pi} = \pi \wedge \pi'$ be defined as $\bar{\pi}_i \triangleq \max\{\pi_i, \pi'_i\}$ and $\underline{\pi}_i \triangleq \min\{\pi_i, \pi'_i\}$ for all $i \in \mathcal{D}$. The following lemma shows that drivers' utilities among all CE outcomes form a lattice, meaning that there exist CE plans where driver utilities are given by $\bar{\pi}$ or $\underline{\pi}$.

The lemma also shows a connection between the top/bottom of the lattice and the welfare differences from losing/replicating a driver, which plays an important role in establishing the incentive properties of the STP mechanism. Denote $W(\mathcal{D}, \mathcal{R})$ as the highest welfare achievable by drivers $\mathcal{D}$ and riders $\mathcal{R}$ (i.e., the optimal objective of (1)). For each driver $i \in \mathcal{D}$, define

the social welfare gain from replicating driver i , and the social welfare loss from losing driver i , as

$$\Phi_{D_i} \triangleq W(\mathcal{D} \cup \{i'\}, \mathcal{R}) - W(\mathcal{D}, \mathcal{R}), \quad (12)$$

$$\Psi_{D_i} \triangleq W(\mathcal{D}, \mathcal{R}) - W(\mathcal{D} \setminus \{i\}, \mathcal{R}), \quad (13)$$

where driver i' with $\theta_{i'} = \theta_i$ is a replica of driver i . A driver-optimal plan has a driver utility profile at the top of the lattice, and a driver-pessimal plan has a utility profile at the bottom of the lattice.

Lemma 4 (Lattice Structure). *Drivers' utility profile π among all CE outcomes form a lattice. Moreover, for each driver $i \in \mathcal{D}$, Φ_{D_i} and Ψ_{D_i} are equal to utility of driver i in the driver-pessimal and driver-optimal CE plans, respectively.*

We leave the proof of this lemma to Online Appendix C.4. The lattice structure follows from the correspondences between driver utilities, the dual LP (7), and the dual of the flow LP, and the fact that optimal dual solutions of MCF form a lattice. Standard arguments on shortest paths in the residual graph (Ahuja et al. 1993) and the connection between optimal dual solutions and subgradients (w.r.t. flow boundary conditions) then imply the correspondence between welfare gains/losses and driver pessimal/optimal utilities.

A plan is in the *core* if no coalition of riders and drivers can break out of this plan and make a plan among themselves, s.t. all drivers and riders in the coalition get at least their utilities from the original plan, and at least one of the drivers or riders is strictly better off.

Lemma 5 (Core Equivalence). *All CE plans are in the core. Moreover, for any budget-balanced core outcome $(x, \tilde{z}, q, r)$, there exists prices p such that the plan with anonymous prices $(x, \tilde{z}, p)$ forms a CE, and has the same driver and rider total utilities.*

See Online Appendix C.5 for the proof of this lemma. Intuitively, any CE plan is in the core because for any $\mathcal{D}' \subseteq \mathcal{D}$ and $\mathcal{R}' \subseteq \mathcal{R}$, the highest achievable coalitional

welfare $W(\mathcal{D}', \mathcal{R}')$ is no greater than the sum of utilities of all driver and riders in this coalition under any CE plan. Given any core outcome, we can construct anonymous trip prices p that support the outcome in CE, and have the same driver and rider total payments: $q_j = x_j p_{o_j, d_j, \tau_j}$ and $r_i = \sum_{j \in \mathcal{R}} \mathbb{1}\{(o_j, d_j, \tau_j, j) \in \tilde{z}_i\} p_{o_j, d_j, \tau_j}$.

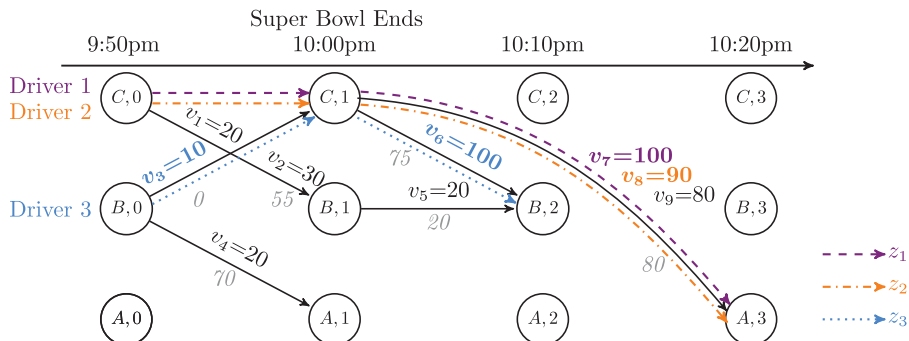
We revisit the Super Bowl example and show that CE plans use prices that are more smooth in space and time, in comparison with the outcome under the myopic pricing mechanism.

Example 2 (Continued). For the Super Bowl example introduced in Section 2, the driver-pessimal CE plan is as shown in Figure 3. All drivers stay at or reposition to location C and pick up riders with high values at time 1. The total rider value is 300, and the total trip costs and exit costs incurred by the drivers are 80 and 5, respectively. This results in an optimal welfare of 215, substantially higher than the welfare of 25 achieved under myopic pricing.

Trip prices are shown in italics below the edges corresponding to the trips. For each feasible path of each driver, the total prices minus costs is 50, which is the welfare gain from replicating the driver (an additional driver at $(C, 0)$ or $(B, 0)$ will be dispatched to $(C, 1)$ and pick up rider 9, improving rider values by 80 and incurring a total cost of 30). The outcome forms a CE that there is no other path with a higher utility for any driver, and all riders are happy with their whether they are picked up given the prices. We can also verify that there is no driver or rider envy and that the outcome is in the core.

In contrast to the myopic pricing mechanism, where the price for the (C, B) trip jumped from 10 to 100 between 9:50 p.m. and 10 p.m., the CE prices started to increase more smoothly *before* the end of the game in anticipation of higher future demand. Intuitively, sending a driver away from location C right before

Figure 3. (Color online) Super Bowl Example: Driver Pessimal Competitive Equilibrium Plan



Notes. Color-coded paths $\{z_i\}_{i \in \mathcal{D}}$ and rider values in bold indicate the movement of drivers in space and time, as well as the riders picked up by each driver. Numbers in italics below trips is the set of driver-pessimal CE prices.

the game ends is costly to the economy, and this is properly reflected in the higher trip prices at 9:50 p.m.

3.3. Static CE Mechanism

Given the existence of welfare-optimal CE plans, we may consider a *static CE mechanism*, which announces a CE plan at time 0, and never again updates the plan even after driver deviations. Rather, each driver can choose to take any feasible path, but can only pick up riders that are dispatched to her and is only paid for the subset of these rider trips that are completed.

Driver best response implies that a static CE mechanism implements a CE plan in dominant strategy; thus, the equilibrium outcome is budget balanced and envy-free. By not updating the plan, however, a static CE mechanism is fragile to driver deviations, which could occur because of mistakes, unexpected contingencies, unexpected traffic, or unmodeled idiosyncratic preferences, and so on. The Super Bowl example demonstrates this lack of robustness: once a driver has deviated, the resulting outcome in the subsequent periods may no longer be reliable or efficient.

Example 2 (Continued). Suppose that driver 3 in the Super Bowl example did not follow the plan at time 0 to pick-up rider 3, but instead stayed in location B until time 1. Under a static CE mechanism with driver-pessimal CE plan (as shown in Figure 3), the effect of this deviation and not updating the plan for the subsequent periods is that driver 3 is no longer able to pick up rider 6 at time 1, who strictly prefers to be picked up given the original price of $p_{C,B,1} = 75$. Driver 2, who was supposed to pick up rider 8 at time 1 is actually able to pick up rider 6 instead of rider 8, and this would lead to a higher welfare. Moreover, driver 3 is now able to pick up rider 5; however, she would not be dispatched to do so.

One may think of a naive fix for this robustness issue of the static CE mechanisms, simply repeating the computation of the plan at all times. However, we show via Examples 5 and 6 in Online Appendix D that such mechanisms fail to be incentive-compatible or envy-free. The challenge is to achieve robustness, but at the same time handle the new strategic considerations that can occur as a result of drivers being able to trigger replanning through deviations.

4. The Spatio-Temporal Pricing Mechanism

In this section, we introduce the STP mechanism and prove our main result, that it is a subgame-perfect equilibrium for drivers to always follow the mechanism's dispatch.

4.1. A Dynamic Mechanism

We first formally define a dynamic ridesharing mechanism that can use the history of actions to update the plan forward from the current state.

Let $s_t = (s_{1,t}, s_{2,t}, \dots, s_{m,t})$ denote the *state* of the ridesharing platform at time t , where each $s_{i,t}$ describes the state of driver $i \in \mathcal{D}$. If driver i has entered the platform and is available at time t at location $a \in \mathcal{L}$, denote $s_{i,t} = (1, a, t)$. Otherwise, if driver i is *en route*, finishing the trip from a to b that she started at time $t' < t$ s.t. $t' + \delta(a, b) > t$, denote $s_{i,t} = (a, b, t')$ if she is relocating with no rider, or $s_{i,t} = (a, b, t', j)$ if she is taking a rider j from a to b at time t' . For drivers that had already exited or decided not to enter, denote $s_{i,t} = \phi$. For drivers with $\underline{\tau}_i \geq t$, that is, who enters or is able to enter now or in the future, $s_{i,t} = (\beta_i, \ell_i, \underline{\tau}_i)$. The initial state of the platform is $s_0 = ((\beta_1, \ell_1, \underline{\tau}_1), \dots, (\beta_m, \ell_m, \underline{\tau}_m))$.

At each time t , each driver i takes an *action* $\alpha_{i,t}$. An available driver i with $s_{i,t} = (1, a, t)$ or $s_{i,t} = (0, a, t)$ may (enter and then) relocate to any location b within reach by the end of the planning horizon (i.e., $b \in \mathcal{L}$ s.t. $t + \delta(a, b) \leq T$), which we denote $\alpha_{i,t} = (a, b, t)$. She may pick up a rider $j \in \mathcal{R}$ with $\tau_j = t$ and $o_j = a$, in which case we write $\alpha_{i,t} = (a, d_j, t, j)$. She may also decide to exit (if $\beta_i = 1$) or not enter (if $\beta_i = 0$), for both cases we denote $\alpha_{i,t} = \phi$. For a driver i that is *en route* at time t , (i.e., $s_{i,t} = (a, b, t')$ or $s_{i,t} = (a, b, t', j)$ for some $t' < t$ s.t. $t' + \delta(a, b) > t$), $\alpha_{i,t} = s_{i,t}$ —the only available action is to finish the current trip. For driver i with $\underline{\tau}_i > t$, denote $\alpha_{i,t} = s_{i,t} = (\beta_i, \underline{\tau}_i, \ell_i)$. A driver with $s_{i,t} = \phi$ takes no more actions: $\alpha_{i,t} = s_{i,t} = \phi$.

The action $\alpha_{i,t}$ taken by driver i at time t determines her state $s_{i,t+1}$ at time $t + 1$:

- (will complete trips at $t + 1$) if $\alpha_{i,t} = (a, b, t')$ or $\alpha_{i,t} = (a, b, t', j)$ s.t. $t' + \delta(a, b) = t + 1$, then $s_{i,t+1} = (1, b, t + 1)$, i.e. becoming available at time $t + 1$ at the destination of their trips,
- (still *en route*) if $\alpha_{i,t} = (a, b, t')$ or $\alpha_{i,t} = (a, b, t', j)$ s.t. $t' + \delta(a, b) > t + 1$, then $s_{i,t+1} = \alpha_{i,t}$,
- (not yet entered) for $i \in \mathcal{D}$ s.t. $\alpha_{i,t} = (\beta_i, \underline{\tau}_i, \ell_i)$, we have $s_{i,t+1} = (\beta_i, \underline{\tau}_i, \ell_i)$, or
- (already exited/never entered) if $\alpha_{i,t} = \phi$, then $s_{i,t+1} = \phi$.

Let $\alpha_t = (\alpha_{1,t}, \alpha_{2,t}, \dots, \alpha_{m,t})$ be the *action profile* of all drivers at time t , and let *history* $h_t \triangleq (s_0, \alpha_0, s_1, \alpha_1, \dots, s_{t-1}, \alpha_{t-1}, s_t)$, with $h_0 = (s_0)$. Finally, let $\mathcal{D}_t(h_t) = \{i \in \mathcal{D} \mid s_{i,t} = (1, a, t) \text{ or } s_{i,t} = (0, a, t) \text{ for some } a \in \mathcal{L}\}$ be the set of drivers available at time t .

Definition 4 (Dynamic Ridesharing Mechanism). A dynamic ridesharing mechanism is defined by its *dispatch rule* α^* , *driver payment rule* r^* , and *rider payment rule* q^* . At each time t , given history h_t and rider information $\mathcal{R}$, the mechanism:

- uses its dispatch rule α^* to determine for each of a subset of available drivers, a *dispatch action* $\alpha_{i,t}^*(h_t)$ to either pick up a rider, or to relocate, or to exit/not enter;

- uses its driver payment rule r^* to determine, for each dispatched driver, a payment $r_{i,t}^*(h_t)$ in the event the driver takes the action ($r_{i,t}^*(h_t) = 0$ for available drivers that are not dispatched);

- dispatches each *en route* driver to keep driving (i.e., $\alpha_{i,t}^*(h_t) = s_{i,t}$) and does not make any payment to driver i in this period: $r_{i,t}^*(h_t) = 0$;

- determines for drivers entering in the future ($i \in \mathcal{D}$ s.t. $\tau_i > t$), and drivers who had already exited ($i \in \mathcal{D}$ s.t. $s_{i,t} = \phi$), $\alpha_{i,t}^*(h_t) = s_{i,t}$ and $r_{i,t}^*(h_t) = 0$; and

- uses its rider payment rule q^* to determine, for each rider who receives a dispatch at time t , the payment $q_j^*(h_t)$ in the event that the rider is picked up.

Each driver then decides on which action $\alpha_{i,t} \in \mathcal{A}_{i,t}(h_t)$ to take, where $\mathcal{A}_{i,t}(h_t)$ is the set of actions available to agent i at time t given history h_t . For an available driver at (a, t) with dispatched action $\alpha_{i,t}^*(h_t)$, $\mathcal{A}_{i,t}(h_t) = \{\alpha_{i,t}^*(h_t)\} \cup \{(a, b, t) \mid b \in \mathcal{L} \text{ s.t. } t + \delta(a, b) \leq T\} \cup \{\phi\}$, that is, the driver can either take the dispatched action, or to relocate to any location, or to exit or not enter; if an available driver at (a, t) is not dispatched, $\alpha_{i,t}^*(h_t)$, $\mathcal{A}_{i,t}(h_t) = \{(a, b, t) \mid b \in \mathcal{L} \text{ s.t. } t + \delta(a, b) \leq T\} \cup \{\phi\}$; for an *en route* driver, or a driver that enters in the future, or a driver that has already exited, $\mathcal{A}_{i,t}(h_t) = \{s_{i,t}\}$. After observing the action profile α_t , the mechanism pays each driver who is dispatched at time t $\hat{r}_{i,t}(\alpha_{i,t}, h_t) = r_{i,t}^*(h_t) \mathbb{1}\{\alpha_{i,t} = \alpha_{i,t}^*\}$, and charges each rider $j \in \mathcal{R}$ with $\tau_j = t$ the amount $\hat{q}_j(\alpha_t) = q_j^*(h_t) \sum_{i \in \mathcal{D}_t} \mathbb{1}\{\alpha_{i,t} = (o_j, d_j, t, j)\}$.

A mechanism is *feasible* if (i) at any time, it is possible for each available driver to take the trip that is dispatched to her, that is, $\forall t, \forall h_t, \forall i \in \mathcal{D}_t$, if $s_{i,t} = (1, a, t)$ or $s_{i,t} = (0, a, t)$ for some $a \in \mathcal{L}$, $\alpha_{i,t}^*(h_t) \in \{(a, b, t) \mid b \in \mathcal{L}, t + \delta(a, b) \leq T\} \cup \{(o_j, d_j, \tau_j, j) \mid j \in \mathcal{R}, \tau_j = t, o_j = a\}$, (ii) no rider is picked up more than once, i.e. $\forall t, \forall h_t, \forall j \in \mathcal{R}$ s.t. $\tau_j = t$, $\sum_{i \in \mathcal{D}_t} \mathbb{1}\{\alpha_{i,t}^*(h_t) = (o_j, d_j, \tau_j, j)\} \leq 1$, and (iii) unavailable drivers are not dispatched. We can see from Definition 4 that there is no payment to or from unavailable or undispached drivers, or a dispatched driver i who deviated from $\alpha_{i,t}^*(h_t)$ at time t , or riders who are not picked up.

Let $\mathcal{H}_t$ be the set of all possible *histories* up to time t . A *strategy* σ_i of driver i defines for all times $t \in [T-1]$ and all histories $h_t \in \mathcal{H}_t$, the action she takes $\alpha_{i,t} = \sigma_i(h_t) \in \mathcal{A}_{i,t}(h_t)$. For a mechanism that always dispatches all available drivers, σ_i^* denotes the *straightforward strategy* of always following the mechanism's dispatches at all times. Let $\sigma = (\sigma_1, \dots, \sigma_m)$ be the *strategy profile*, with $\sigma_{-i} = (\sigma_1, \dots, \sigma_{i-1}, \sigma_{i+1}, \dots, \sigma_m)$. The strategy profile σ , together with the initial state s_0 and the rules of a mechanism, determine all actions and payments of all drivers through the planning horizon. Let $\sigma_i|_{h_t}$, $\sigma|_{h_t}$ and $\sigma_{-i}|_{h_t}$ denote the strategy profile from time t and history h_t onward for driver i , all drivers, and all drivers but i , respectively.

For each $j \in \mathcal{R}$, let $\hat{x}_j(\sigma) \in \{0, 1\}$ be the indicator that rider j is picked up given strategy σ , and let $\hat{q}_j(\sigma) = \hat{x}_j(\sigma)q_j^*(h_{\tau_j})$ be her actual payment. For each driver $i \in \mathcal{D}$, $\hat{r}_i(\sigma) \triangleq \sum_{t=0}^{T-1} \hat{r}_{i,t}(\sigma_i(h_t), h_t)$ denotes the total actual payments made to driver i , where drivers follow σ and the history h_t is induced by the initial state and strategy σ . Let $\hat{\pi}_{i,t}(\sigma_i(h_t), h_t)$ be the actual utility driver i gets at time t given history h_t and strategy σ_i . We know that if $\sigma_i(h_t) = (a, b, t)$ or $\sigma_i(h_t) = (a, b, t, j)$, then $\hat{\pi}_{i,t}(\sigma_i(h_t), h_t) = \hat{r}_{i,t}(\sigma_i(h_t), h_t) - c_{a,b,t}$; if $\sigma_i(h_t) = \phi$ and $s_{i,t} = (1, a, t)$ for some $a \in \mathcal{L}$, then $\hat{\pi}_{i,t}(\sigma_i(h_t), h_t) = \hat{r}_{i,t}(\sigma_i(h_t), h_t) - \kappa_{T-t}$. For every other scenario, $\hat{\pi}_{i,t}(\sigma_i(h_t), h_t) = \hat{r}_{i,t}(\sigma_i(h_t), h_t)$. Denote $\hat{\pi}_i(\sigma) \triangleq \sum_{t=0}^{T-1} \hat{\pi}_{i,t}(\sigma_i(h_t), h_t)$ as driver i 's total utility.

Fixing driver and rider types, a ridesharing mechanism induces a finite horizon extensive form game. At each time point t , each driver decides on an action $\alpha_{i,t} = \sigma_i(h_t) \in \mathcal{A}_{i,t}(h_t)$ to take based on strategy σ_i and the history h_t , and receives utility $\hat{\pi}_{i,t}(\alpha_{i,t}, h_t)$. The total utility $\hat{\pi}_i(\sigma)$ to each driver is determined by the rules of the mechanism and the strategy profile *strategy*.

We define the following properties.

Definition 5 (Budget Balance). A ridesharing mechanism is *budget balanced* if for any set of riders and drivers, and any strategy profile σ taken by the drivers, we have $\sum_{j \in \mathcal{R}} \hat{q}_j(\sigma) \geq \sum_{i \in \mathcal{D}} \hat{r}_i(\sigma)$.

Definition 6 (Subgame-Perfect Incentive Compatibility). A ridesharing mechanism that always dispatches all available drivers is *subgame-perfect incentive compatible* (SPIC) for drivers if given any set of riders and drivers, following the mechanism's dispatches at all times forms a subgame-perfect equilibrium (SPE) among drivers, meaning for all $t \in [T-1]$, for any $h_t \in \mathcal{H}_t$,

$$\sum_{t'=t}^{T-1} \hat{\pi}_{i,t'}(\sigma_i^*|_{h_t}, \sigma_{-i}^*|_{h_t}) \geq \sum_{t'=t}^{T-1} \hat{\pi}_{i,t'}(\sigma_i|_{h_t}, \sigma_{-i}^*|_{h_t}), \quad \forall \sigma_i, \quad \forall i \in \mathcal{D}. \quad (14)$$

A ridesharing mechanism is *dominant strategy incentive compatible* (DSIC) if for any driver, following the mechanism's dispatches at all times maximizes her total utility, regardless of the actions taken by the rest of the drivers.

Definition 7 (Individual Rationality (IR)). A ridesharing mechanism that always dispatches all available drivers is *individually rational in SPE for drivers* if for any set of riders and drivers, (i) the mechanism is SPIC for drivers, and (ii) assuming σ^* , drivers that have not yet entered do not get negative utility from participating, that is, $\hat{\pi}_i(\sigma^*) \geq 0$ for all $i \in \mathcal{D}$ s.t. $\beta_i = 0$. A ridesharing mechanism is *individually rational for riders* if for any set of riders and drivers, and any

strategy profile σ taken by the drivers, $\hat{x}_j(\sigma)v_j \geq \hat{q}_j(\sigma)$ for all riders $i \in \mathcal{R}$.

Definition 8 (Envy-Freeness in SPE). A ridesharing mechanism that always dispatches all available drivers is *envy-free in SPE for drivers* if for any set of riders and drivers, (i) the mechanism is SPIC for drivers, and (ii) for any time $t \in [T-1]$, for all history $h_t \in \mathcal{H}_t$, all drivers with the same state at time t have the same continuation payoff in the subsequent periods, assuming all drivers follow the mechanism's dispatches: $\sum_{i'=t}^{T-1} \hat{\pi}_{i',t'}(\sigma^*|h_t) = \sum_{i'=t}^{T-1} \hat{\pi}_{i',t'}(\sigma^*|h_t)$, $\forall i, i' \in \mathcal{D}$ s.t. $s_{i,t} = s_{i',t}$. A ridesharing mechanism is *envy-free in SPE for riders* if (i) the mechanism is SPIC for drivers, and (ii) for all $j \in \mathcal{R}$, for all possible $h_{\tau_j} \in \mathcal{H}_{\tau_j}$, and all $j' \in \mathcal{R}$ s.t. $(o_j, d_j, \tau_j) = (o_{j'}, d_{j'}, \tau_{j'})$, $\hat{x}_j(\sigma^*)v_j - \hat{q}_j(\sigma^*) \geq \hat{x}_{j'}(\sigma^*)v_{j'} - \hat{q}_{j'}(\sigma^*)$.

Definition 9 (Core-Selecting). A ridesharing mechanism that always dispatches all available drivers is *core-selecting* if for any set of riders and drivers, (i) the mechanism is SPIC, and (ii) for all time $t \in [T-1]$ and any history $\forall h_t \in \mathcal{H}_t$ onward, the outcome under σ^* is in the core.

Fix a mechanism with dispatch rule α^* and payment rules q^*, r^* , where all available drivers are always dispatched. Recall that the outcome under the straightforward strategy σ^* over the entire planning horizon can be computed at time 0, and is called the *time 0 plan* of the mechanism. If some driver deviated at time $t-1$ for some $t > 0$, the downstream outcome given the dispatching and payment rules, assuming all drivers follow $\sigma^*|_{h_t}$, can be thought of as an updated time t plan.

For any time $t \in [T]$, given any state s_t of the platform, let $E^{(t)}(s_t)$ represent the *time-shifted economy* starting at state s_t , with planning horizon $T^{(t)} = T - t$, the same set of locations $\mathcal{L}$ and distances δ , and the remaining riders $\mathcal{R}^{(t)} = \{(o_j, d_j, \tau_j - t, v_j) \mid j \in \mathcal{R}, \tau_j \geq t\}$. For drivers, we have $\mathcal{D}^{(t)}(s_t) = \{\theta_i^{(t)} \mid i \in \mathcal{D}\}$, with types $\theta_i^{(t)} = (\beta_i^{(t)}, \ell_i^{(t)}, \underline{\tau}_i^{(t)}, \bar{\tau}_i^{(t)})$ determined as follows:

- for available drivers $i \in \mathcal{D}$ s.t. $s_{i,t} = (1, a, t)$ or $s_{i,t} = (0, a, t)$ for some $a \in \mathcal{L}$, let $\theta_i^{(t)} = (\beta_i^{(t)}, \ell_i^{(t)}, \underline{\tau}_i^{(t)}, \bar{\tau}_i^{(t)}) = (1, a, 0, \bar{\tau}_i - t)$ or $(1, a, 0, \bar{\tau}_i - t)$, respectively,
- for en route drivers $i \in \mathcal{D}$ s.t. $s_{i,t} = (a, b, t')$ or (a, b, t', j) where $t' + \delta(a, b) > t$, let $\theta_i^{(t)} = (\beta_i^{(t)}, \ell_i^{(t)}, \underline{\tau}_i^{(t)}, \bar{\tau}_i^{(t)}) = (1, b, t' + \delta(a, b) - t, \bar{\tau}_i - t)$,
- for each driver $i \in \mathcal{D}$ with $\underline{\tau}_i > t$, let $\theta_i^{(t)} = (\beta_i^{(t)}, \ell_i^{(t)}, \underline{\tau}_i^{(t)}, \bar{\tau}_i^{(t)}) = (\beta_i, \ell_i, \underline{\tau}_i - t, \bar{\tau}_i - t)$, and
- exclude drivers that have already exited or chose not to enter.

Definition 10 (Temporal Consistency). A ridesharing mechanism is *temporally consistent* if after deviation at time $t-1$, the updated plan is identical to that determined for economy $E^{(t)}(s_t)$.

Upon deviation(s) at time $t-1$, a temporally consistent mechanism updates its plan from time t onward as if t is the beginning of the planning horizon and thus does not make use of time-extended contracts. A temporally inconsistent mechanism can trivially align incentives by firing any driver who has deviated or by threatening to “shut down” after any deviation, for example.

4.2. The Spatio-Temporal Pricing Mechanism

We define the STP mechanism by providing a method to plan or replan, thereby implicitly defining the dispatch and payment rules. For each $a \in \mathcal{L}$ and $t \in [T]$, denote the welfare gain from an additional driver at (a, t) that is already in the platform as

$$\Phi_{a,t} \triangleq W(\mathcal{D} \cup \{(1, a, t, T)\}) - W(\mathcal{D}), \quad (15)$$

where $(1, a, t, T)$ represents the type of this driver that stays until the end of the planning horizon.

Definition 11 (The STP Mechanism). The *STP mechanism* is a dynamic ridesharing mechanism that always dispatches all available drivers. Given economy $E^{(0)}$ at the beginning of the planning horizon or economy $E^{(t)}(s_t)$ immediately after a deviation by one or more drivers, the mechanism completes the following planning step:

- *Dispatch rule*: To determine the dispatches (α^*), compute an optimal solution (x, y) to the LP (1), and dispatch each driver i to take the path $Z_{i,k}$ for k s.t. $y_{i,k} = 1$, and pick up riders with $x_j = 1$,

- *Payment rules*: To determine driver and rider payments (r^* and q^*), for each $(a, b, t) \in \mathcal{T}$, set anonymous trip prices to be

$$p_{a,b,t} = \Phi_{a,t} - \Phi_{b,t+\delta(a,b)} + c_{a,b,t}. \quad (16)$$

For each rider $j \in \mathcal{R}$, $q_j^* = p_{o_j, d_j, \tau_j} \sum_{i \in \mathcal{D}} \mathbb{1}\{\alpha_{i, \tau_j}^* = (o_j, d_j, \tau_j, j)\}$. For each driver $i \in \mathcal{D}$, $r_{i,t}^* = \sum_{j \in \mathcal{R}, \tau_j = t} p_{o_j, d_j, t} \mathbb{1}\{\alpha_{i, \tau_j}^* = (o_j, d_j, t, j)\}$.

We now state the main result of the present paper.

Theorem 1. *The STP mechanism is temporally consistent and subgame-perfect incentive compatible. It is also individually rational for riders and strictly budget balanced for any action profile taken by the drivers. From any history onward, the equilibrium outcome is welfare-optimal, core-selecting, envy-free, and individually rational for drivers.*

The proof of Theorem 1 is provided in Online Appendix C.6. We first show that the total utility of each driver under the STP mechanism is $\pi_i = \Phi_{D_i}$, the welfare gain from replicating driver i . Setting $u_j = \max\{v_j - p_{o_j, d_j, \tau_j}, 0\}$ for all $j \in \mathcal{R}$, we show that (p, π, u) forms an optimal solution to the dual LP (7) by observing (i) (Φ, u) forms an optimal solution to the dual of the corresponding MCF problem (the proof of Lemma 4), and (ii) a correspondence between the

optimal solutions of the dual LP (7) and the optimal solutions of the dual of the MCF (Lemma 7 in Online Appendix C.3). This implies that the plan determined by the STP mechanism starting from any history onward forms a CE, and as a result is individually rational, budget balanced, envy-free, and resides in the core.

For incentive alignment, the single-deviation principle (Osborne and Rubinstein 1994) implies that we only need show that a single deviation from the dispatching is not useful. For any driver available at some location a and time t , her total utility from time t onward, if all drivers follow the dispatches, is equal the welfare gain (at the time when the plan is computed) from adding an extra driver at location a and time t . We establish that this welfare gain is weakly higher than the welfare gain for the economy (at time $t + 1$) from replicating this driver at any location and time that the driver can deviate and relocate to.

For this, we use the M^h concavity (and more specifically, the local exchange properties) of optimal objectives of MCF problems (Murota 2003). In particular, to maximize welfare, there is stronger substitution among drivers at the same location and time, than among drivers at different locations or times. This shows that declining the mechanism's dispatch to stay/relocate is not useful. We also show that none of (i) exiting earlier than dispatched, (ii) not entering/exiting when asked to, and (iii) entering when dispatched not to, is a useful deviation.

Under the STP mechanism, replanning can be triggered by the deviation of any driver; thus, the utility of a driver is affected by the actions of others and the mechanism is not DSIC. We also show in the following theorem that no mechanism can implement the desired properties in a dominant-strategy equilibrium. See Online Appendix C.7 for the proof of the theorem.

Theorem 2. *Following the mechanism's dispatch at all times does not form a dominant strategy equilibrium under any dynamic ridesharing mechanism that is, from any*

history onward, (i) welfare-optimal, (ii) IR for riders, (iii) budget balanced, and (iv) envy-free for riders and drivers.

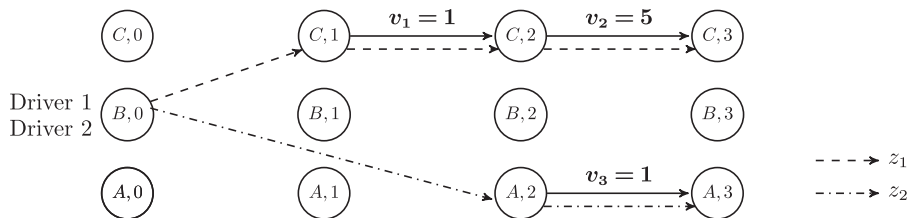
A natural variation on the STP mechanism is the driver-optimal analog, which always computes a driver-optimal CE plan at the beginning of the planning horizon or upon the deviation of any driver. Under this mechanism, a driver's continuation payoff from some location and time onward is equal to her "marginal product," that is, the welfare loss in the economy from losing a driver at this location and time. Despite that it reflects the payments of a VCG mechanism, the following example shows that the driver-optimal mechanism is not incentive compatible. This is because a driver's marginal product can increase over time, as the set of trips that can be completed by the rest of the drivers becomes smaller. In these scenarios, the driver may deviate from the mechanism's dispatch, trigger a recomputation of the downstream plan, and get paid the updated, higher marginal product in the subsequent periods.

Example 3. Consider the economy illustrated in Figure 4, with three locations, three time periods, and symmetric distances $\delta(A,A) = \delta(B,B) = \delta(C,C) = \delta(B,C) = 1$, $\delta(A,B) = \delta(A,C) = 2$. All trip costs and early exit costs are zero. Two drivers enter the platform at time 0 at location B , and three riders have types:

- Rider 1: $o_1 = C$, $d_1 = C$, $\tau_1 = 1$, $v_1 = 1$,
- Rider 2: $o_2 = C$, $d_2 = C$, $\tau_2 = 2$, $v_2 = 5$,
- Rider 3: $o_3 = A$, $d_3 = A$, $\tau_3 = 2$, $v_3 = 1$.

In a welfare-optimal dispatching as shown in Figure 4, driver 1 is dispatched to take the path $z_1 = ((B,C,0), (C,C,1), (C,C,2))$ and to pick up riders 1 and 2. Driver 2 takes the path $z_2 = ((B,A,0), (A,A,2))$ and picks up rider 3. Every driver-optimal CE plan sets anonymous trip prices $p_{A,A,2} = 1$ and $p_{C,C,1} + p_{C,C,2} = 1$, so that the total utility of each driver is 1, which is the welfare loss if one driver is removed from the economy.

Assume that driver 2 follows the mechanism's dispatch and starts to drive toward location A at time 0, we show a useful deviation of driver 1 by rejecting the



Note. Under a welfare-optimal plan, driver 1 picks up riders 1 and 2 and driver 2 picks up rider 3.

dispatched relocation to C and staying in location B . At time 1, when the platform updates the downstream plan, driver 2 is already *en route* to A and thus the only rider she can pick up in the future is rider 3. Driver 1 would be asked to relocate to C and then pick up rider 2. The price $p_{C,C,2}$ in the updated driver-optimal CE plan would be 5, which is the welfare loss if the economy at time 1 loses driver 1 at time 1. This is higher than driver 1's payment from following the dispatches at all times.

This kind of useful deviation does not exist under the STP mechanism, because under the driver-pessimal CE plan, each driver is paid the additional welfare the economy gains if we replicate this driver. It is always more useful to have an extra driver earlier, and thus the “replica welfare gain” is monotonically nonincreasing over time.

A variation on the driver-optimal mechanism where drivers' payments are shifted in time is equivalent to the *dynamic VCG mechanism* (Cavallo et al. 2009, Bergemann and Välimäki 2010). The dynamic VCG mechanism does not align incentives for drivers either, because the existence of a driver at some point of time may exert negative externality on the economy, in which case the payment to the driver would be negative for that time period. The driver would have incentives to decline the dispatch and avoid such payment. See discussions and examples in Online Appendix E.1. The VCG mechanisms would address information asymmetry, but these examples highlight that the challenge we face is rather one of aligning incentives in the absence of time-extended contracts.

5. Concluding Remarks

We study the problem of pricing and dispatching in ridesharing platforms in a way that drivers would always choose to accept the platform's dispatches. Under a complete information, discrete time, multiperiod and multilocation model, we show that always accepting the dispatched trips forms a subgame-perfect equilibrium among the drivers under the STP mechanism. The STP mechanism is simple in design: computing a driver-pessimal competitive equilibrium plan at the beginning of the planning horizon, as well as after any deviation from this plan.

The main operational insight from this paper, as discussed in the introduction, is the role of the marginal welfare contribution of adding an extra driver at the origin versus at the destination of a trip (as prescribed in (16)), in determining prices that are appropriately smooth and optimizing social welfare.

Welfare vs. Revenue Optimization. We focus on welfare instead of revenue optimization in this paper. With the substantial network effect in ridesharing marketplaces, as well as fierce competition, major platforms such as Uber and Lyft have been prioritizing

growth instead of profit. Welfare optimization is therefore aligned with this goal and should also be the objective when a city or a nonprofit organization (e.g., RideAustin) is operating a ridesharing platform.

In regard to revenue, the STP mechanism strictly balances budget. Alternatively, we many think about the platform taking a fixed fraction of the driver surplus. This does not affect the results presented in this paper, and the outcome under STP still resides in some ϵ -core of the market, depending on the cut taken by the platform. (The (multiplicative) ϵ -core of a game is the set of outcomes where there does not exist a coalition of participants who can form an alternative plan among themselves, such that all participants' utility increases by a fraction of ϵ .)

Modeling Assumptions. Throughout the paper, we assume that drivers have homogeneous costs and will all stay until the end of the planning horizon and that riders are impatient. Examples 7 and 8 in Online Appendix D.2 show that if either assumption is relaxed, the LP relaxation (1) for the optimal planning problem may no longer be integral because the reduction to the MCF problem fails. As a result, there may not exist anonymous, origin-destination CE prices.

As the length of a time period becomes smaller, it is less likely for riders to be fully impatient, although prices that are appropriately smooth in time would reduce riders' incentives to strategically wait for a lower price. As a time period gets shorter, the envy-freeness property also weakens—the “equivalence of rights” depends on time and location, and there would be fewer riders requesting the same trip or drivers at the same location at the same time.

Complete Information. We assume complete information and a finite planning horizon throughout this paper. A natural next step is to generalize the model to settings where there is uncertainty about supply and demand, and where the planning horizon rolls forward as the uncertainty resolves over time. One challenge is that the class of $M^\sharp$ concave functions is not closed under addition; thus, the continuation value (i.e., the expected future welfare as a function of the positioning of drivers at the end of this immediate planning period) may no longer be $M^\sharp$ concave. This may affect the integrality of the optimal planning problem and the existence of CE. Conversely, although we have made use of CE to establish envy-freeness and core-selecting properties, the use of CE prices is not necessary for incentive alignment, because a driver cannot simply pick a series of trips to complete, so this does not by itself prevent progress. Building predictive models for supply and demand in ridesharing platforms has proved challenging, and it may be more promising and practical to directly estimate the welfare contributions of an extra driver using existing data from the current operations.

Information Elicitation. The focus of this paper is incentive alignment in a dynamic environment while maintaining driver flexibility and not on information asymmetry. This said, we do have some results in this regard. On the driver's side, we may consider a scenario where the mechanism does not have the driver entrance information but asks each driver to report at the beginning of the planning horizon the time and location at which she will enter. We show in Online Appendix D.4 that a driver with entrance location and time $(\ell_i, \underline{\tau}_i)$ has no incentive to report some feasible location and time $(\hat{\ell}_i, \hat{\tau}_i)$ where $\hat{\tau}_i \geq \underline{\tau}_i + \delta(\ell_i, \hat{\ell}_i)$, and then actually enter the platform at $(\hat{\ell}_i, \hat{\tau}_i)$.

On the rider side, Example 9 in Online Appendix D.3 shows that the STP mechanism is not truthful for riders. This is a departure from the classical, unit-demand assignment problem (Shapley and Shubik 1971), where the seller-pessimal outcome corresponds to the buyer-side VCG prices and is truthful for buyers. We show in Theorem 4 that the rider-side VCG payment for a rider is equal to the minimum price for her trip among all CE outcomes, but such trip prices, when adopted for all riders, may not form a CE (i.e., the anonymous trip prices among all CE outcomes does not form a lattice). This implies that no budget-balanced core-selecting mechanism is truthful for riders, and we also show that no budget-balanced, optimal and subgame-perfect incentive compatible mechanism is truthful for riders.

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